

Blake Allen Linders

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Education

Texas A&M University

B.S. Electrical Engineering, Minor in Mathematics - GPA: 3.51

August 2024 - Present (Expected May 2028)

College Station, TX

- **Relevant Coursework:** Electrical Circuit Theory, Digital Systems Design, Machine Learning for EE, Program Design & Concepts in C++, Electricity & Magnetism, Differential Equations, Applied Mathematics
- **Current Coursework:** Electric Energy Conversion, Signals & Systems, Electronics, Electric & Magnetic Fields
- **Certifications:** Data Analysis Certificate - Katzberg Institute
- **Involvement:** IEEE Member (including meetings and workshops) and Multiple Intramural Sports

Skills

Electrical: AC/DC and phasor analysis, RC transients, Sallen-Key active filters, op-amp oscillators, magnitude/phase response, breadboard prototyping, PCB schematic capture and placement

Digital/Embedded: Verilog, Moore/Mealy FSMs, clock division, switch debouncing, FPGA implementation, ESP32, PWM, WebSockets

Machine Learning/Data: PyTorch, scikit-learn, CNNs, RNNs, regression, classification, pandas, NumPy, Plotly

Tools/Languages: Oscilloscope, multimeter, DAQ, LTspice, KiCad, Vivado, MATLAB/Simulink; Python, C++, SQL, JavaScript, HTML/CSS, Git/GitHub

Projects

ESP32 Three-Wheel RC Vehicle - Independent Project

May 2026 - Present

- Built and troubleshot a breadboard prototype integrating an ESP32, L293D motor driver, ultrasonic sensor, and mixed-voltage circuitry; resolved insufficient motor power and designed a two-layer KiCad PCB incorporating Schottky-diode protection and power-supply decoupling.
- Developed a browser-based WebSocket controller hosted through an ESP32 SoftAP and captive portal, translating simultaneous touch inputs into PWM motor commands with a 500 ms communication-loss safety stop.

AC Circuit Analysis and Active Filter Validation - ECEN 214 Laboratory

January 2026 - May 2026

- Collaboratively built first-, second-, and cascaded third-order circuits and measured magnitude and phase response from 100 Hz to 10 kHz using a signal generator and oscilloscope; compared hardware results with theoretical transfer functions.
- Characterized RC, voltage-controlled oscillator, and underdamped Sallen-Key behavior; estimated a nominal 100 nF capacitor at 94.4 nF average and measured damping and frequency values within approximately 5% of calculations.

FPGA Digital Systems & Hardware Validation - ECEN 248 Laboratory

August 2025 - December 2025

- Implemented Verilog counters, clock dividers, and switch-debouncing logic; verified behavior through Vivado waveforms and physical deployment on a ZYBO Z7-10 FPGA.
- Designed structural/dataflow adders and Moore/Mealy FSMs, modeling approximately 9 ns delay for a 16-bit carry-lookahead design versus roughly 50 ns for ripple-carry and deploying combination-lock and six-state traffic-control systems.

Experience & Leadership

AggiE-Challenge Research Abroad

September 2026 - December 2026

AI-Enhanced Feedback Control Systems - ENGR 491 Research

Paris, France

- Selected for an upcoming research project to model a benchmark feedback system, implement an ML/RL-based controller, compare its performance with classical control, and present the results in France.

Global Career Accelerator

May 2026 - August 2026

Coding for Data Academic Intern - UGST 484 Internship

Remote

- Queried a 601,740-record synthetic Intel sustainability dataset using joins, CTEs, CASE logic, aggregations, and window functions to compare energy and emissions impacts by device age, type, and region.
- Used pandas and Plotly to analyze multi-year Grammy and Recording Academy web data, quantify 43x awards-night traffic, compare engagement metrics, and deliver a data-supported business recommendation.

Blue Stars Drum and Bugle Corps

May 2025 - August 2025

Section Leader & Performing Member, Bassline

La Crosse, WI

- Led four bassline members and co-led 11 additional percussion performers through 12-hour rehearsal days on a national performance tour spanning 50+ cities and 30+ performances; coordinated schedules and rehearsal strategy with instructional staff.
- Received the Outstanding First Year Battery Member Award.

Lewisville High School

August 2024 - April 2025

Drumline Technician

Lewisville, TX

- Instructed five bassline students and co-taught eight additional percussionists, developing fundamentals and improving ensemble sound.

Pacific Crest Youth Arts Organization

May 2024 - August 2024

Performing Member, Bassline

Diamond Bar, CA

- Completed a national performance tour requiring 12-hour rehearsal days, 50+ cities, and 30+ performances.